

EDUCATION

UC Irvine • B.S. in Computer Science w/ Honors, 3.9 / 4.0 GPA Sep 2024 – Jun 2026

- graduated in 2 years
- ICS Honors Program
- UROP Fellow
- Data Structures
- Algorithms
- Software Eng.
- Compilers
- Cryptography
- Formal Lang. & Automata
- Quantum Computation
- Distributed Systems

PROJECTS

tatic • *Rust, WebAssembly, dependent type theory* Mar 2026 – Present

- Built a **JIT** for a higher-order language, caching compiled **WebAssembly** by content hash.
- Checks compiled code against an interpreter and, where possible, a **kernel-checked proof**.

Lambast • *Honors research project using Rust* Feb 2025 – Jun 2026

- Built a whole-program optimizing compiler for untyped λ -calculus using **closure conversion** and **De Bruijn indices**.
- Generated randomized benchmark terms via Grygiel & Lescanne's binary lambda-term enumeration; presented at the **UCI UROP Symposium**.

renno • *Rust* Sep 2026 – Present

- Built a scripting language on a **CEK machine** with multi-shot **algebraic effect handlers**.
- Added **gradual typing**, row-polymorphic effect checking, and refinement predicates; 426 tests.

cereal • *C99* Sep 2026 – Present

- Wrote a C99/GNU front end in C99: GCC-compatible preprocessor with macro lint, parser, and checker.
- Matches gcc-13 diagnostics on over 96% of gcc.dg; parallel -E is fuzzed for identical output.

SKILLS AND INTERESTS

Proficient in Rust, C / C++, JS / TS, Python, Svelte, Tailwind.

Familiar with Git, HTML / CSS, Nix, LaTeX, Typst, Claude Code, Unix systems & shells.

Acquainted with most mainstream languages, e.g., C#, Go, Haskell, Java, Lua, Ruby, SQL.

Research interests $\approx$ programming language design & theory.

EXPERIENCE

Learning Assistant • UC Irvine Mar 2025 – Jun 2026

- Delivered supplementary pedagogy for **Python** and upper-div **Algorithms**, e.g., Master Theorem.
- Spearheaded study sessions to identify and guide students through gaps in understanding.

Intern • LGBTQIA+ Identity Commission @ UC Irvine Oct 2024 – Jun 2025

- Led a team of 3 to organize and host themed Gayme Show™.

Programmer • Video Game Development Club @ UC Irvine Sep 2024 – May 2025

- Worked with team of 13 to create typing game in **C#** and **Unity**; led dev of the main UI window.